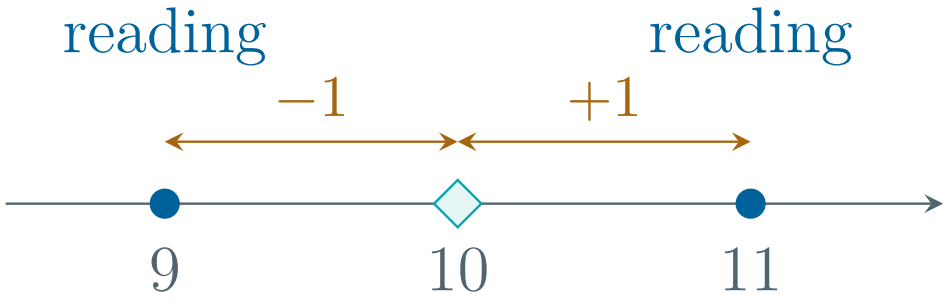
Scalar illustration: equal precision



estimate $\hat{x} = 10$

Network weighted least squares

$$\hat{x} = \arg\min_x \sum_j \left(\frac{z_j - (Hx)_j}{\sigma_j} \right)^2$$

z_j: reading; $(Hx)_j$: prediction

σ_j: stated noise scale